

$$y(t) = c_1 \cos \omega t + c_2 \sin \omega t$$

~~$$A \vec{x} = \vec{b}$$~~

$$\begin{bmatrix} c_1 & c_2 \end{bmatrix} \begin{bmatrix} \cos \omega t \\ \sin \omega t \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

Closure under addition

$$f(t) = a_1 \cos \omega t + a_2 \sin \omega t$$

$$g(t) = b_1 \cos \omega t + b_2 \sin \omega t$$

$$f(t) + g(t) = a_1 \cos \omega t + a_2 \sin \omega t + b_1 \cos \omega t + b_2 \sin \omega t$$

$$= (a_1 + b_1) \cos \omega t + (a_2 + b_2) \sin \omega t$$

$$a_1, a_2, b_1, b_2 \in \mathbb{R} \quad b_1, a_2 + b_2 \in \mathbb{R}$$

$$(a_1 + b_1), (a_2 + b_2) \in \mathbb{R}$$

Closure under scalar multiplication

$$f(t) = a_1 \cos \omega t + a_2 \sin \omega t$$

$$c \cdot f(t) = c(a_1 \cos \omega t + a_2 \sin \omega t)$$

$$= ca_1 \cos \omega t + ca_2 \sin \omega t$$

$$a_1, a_2 \in \mathbb{R} \quad ca_1, ca_2 \in \mathbb{R}$$

Vector Space definition, axioms

1. $\vec{u} + \vec{v}$ is in V
2. $\vec{u} + \vec{v} = \vec{v} + \vec{u}$
3. $(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$
4. There is a zero vector, $\vec{u} + \vec{0} = \vec{u}$
5. For each $\vec{u}$ in V , there is a $-\vec{u}$ such that $\vec{u} + (-\vec{u}) = \vec{0}$
6. $c\vec{u}$ is in V (scalar multiplication)
7. $c(\vec{u} + \vec{v}) = c\vec{u} + c\vec{v}$
8. $(c+d)\vec{u} = c\vec{u} + d\vec{u}$
9. $c(d\vec{u}) = (cd)\vec{u}$
10. $1\vec{u} = \vec{u}$

Presence of zero vector

By selecting $c_1 = c_2 = 0$, we can obtain the zero vector for any t .

3800 Introducing Column Space (video)

Two subspaces of $\mathbb{R}^n$

$$A_{m \times n} = \begin{array}{|c|c|} \hline a_{11} & a_{1n} \\ \hline \vdots & \vdots \\ \hline a_{m1} & a_{mn} \\ \hline \end{array} \begin{array}{l} \text{row 1 } \vec{r}_1 \\ \vdots \\ \text{row } m \vec{r}_m \end{array}$$

$\vec{v}_1 \qquad \qquad \vec{v}_n$

Each column vector $\vec{v}_x$, has the same number of entries as there are rows, m .

$$\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n \in \mathbb{R}^m$$

$$\vec{r}_1, \vec{r}_2, \dots, \vec{r}_m \in \mathbb{R}^n$$

Column space of A is the set of all linear combinations of the columns in A .

$$\text{Col } A_{m \times n} = \text{Span}\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$$

Column Space Definition

The column space of a matrix A is the set $\text{Col } A$ of all linear combinations of the columns of A .

The column space of a matrix A is the set $\text{Col } A$ of all linear combinations of the columns of A .

If $A = [\vec{a}_1 \dots \vec{a}_n]$ with columns in $\mathbb{R}^m$, then $\text{Col } A$ is the same as $\text{span}\{\vec{a}_1, \dots, \vec{a}_n\}$. Example 4 shows that the column space of an $m \times n$ matrix is a subspace of $\mathbb{R}^m$. Note that $\text{Col } A$ equals $\mathbb{R}^m$ only when the columns of A span $\mathbb{R}^m$. Otherwise, $\text{Col } A$ is only part of $\mathbb{R}^m$.

Let $A = \begin{bmatrix} 1 & -3 & -4 \\ -4 & 6 & -2 \\ -3 & 7 & 6 \end{bmatrix}$, $\vec{b} = \begin{bmatrix} 3 \\ 3 \\ -4 \end{bmatrix}$

If $\vec{b}$ is a linear combination of the columns in A , then $\vec{b} \in \text{Col } A$.

$$\left[\begin{array}{ccc|c} 1 & -3 & -4 & 3 \\ -4 & 6 & -2 & 3 \\ -3 & 7 & 6 & -4 \end{array} \right]$$

$$R_2 \leftarrow R_2 + 4R_1$$

$$R_3 \leftarrow R_3 + 3R_1$$

$$\left[\begin{array}{ccc|c} 1 & -3 & -4 & 3 \\ 0 & -6 & -18 & 15 \\ 0 & -2 & -6 & 5 \end{array} \right]$$

$$R_3 \leftarrow R_3 + \frac{1}{3}R_2$$

$$\left[\begin{array}{ccc|c} 1 & -3 & -4 & 3 \\ 0 & -6 & -18 & 15 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$\rightarrow \vec{b}$ is linearly dependent

$$R_2 \leftarrow R_2 - (-1/6)R_1$$

$$\left[\begin{array}{ccc|c} 1 & -3 & -4 & 3 \\ 0 & 1 & 3 & 5/2 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$R_1 \leftarrow R_1 + 3R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 5 & -9/2 \\ 0 & 1 & 3 & 5/2 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Let's check

$$\vec{b} = -9/2 \begin{bmatrix} 1 \\ -4 \\ -3 \end{bmatrix} + (-5/2) \begin{bmatrix} -3 \\ 6 \\ 7 \end{bmatrix} = \begin{bmatrix} -9/2 \\ 18 \\ 27/2 \end{bmatrix} + \begin{bmatrix} 15/2 \\ -15 \\ -35/2 \end{bmatrix}$$

$$\vec{b} = \begin{bmatrix} 3 \\ 3 \\ -4 \end{bmatrix}$$

3800 Example and theorem column space (video)

Example with column space

$$\begin{aligned} x_1 + 3x_2 &= -2 & \text{Is this solvable? - Yes!} \\ 4x_1 - 6x_2 &= 10 & \text{And what does this mean?} \end{aligned}$$

Solution: span subspace

$$\left[\begin{array}{cc|c} 1 & 3 & -2 \\ 4 & -6 & 10 \end{array} \right] \quad A\vec{x} = \vec{b} = \begin{bmatrix} 1 & 3 \\ 4 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -2 \\ 10 \end{bmatrix}$$

In column form we can say

$$x_1 \begin{bmatrix} 1 \\ 4 \end{bmatrix} + x_2 \begin{bmatrix} 3 \\ -6 \end{bmatrix} = \begin{bmatrix} -2 \\ 10 \end{bmatrix}$$

$\vec{v}_1 \quad \vec{v}_2$

which is the same as saying that:
are there any scalars x_1 and x_2
that will make $\vec{b}$ a linear
combination of the column
vectors in A ?

$$[A|\vec{b}] = \left[\begin{array}{cc|c} 1 & 3 & -2 \\ 4 & -6 & 10 \end{array} \right] \quad R_2 \leftarrow R_2 - 2R_1$$
$$\left[\begin{array}{cc|c} 1 & 3 & -2 \\ 0 & -9 & 9 \end{array} \right]$$

$$R_2 \leftarrow R_2 \cdot \frac{1}{9}$$

$$\left[\begin{array}{cc|c} 1 & 3 & -2 \\ 0 & -1 & 1 \end{array} \right]$$

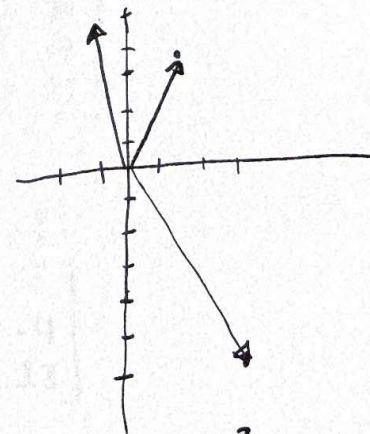
span the subspace

$$\left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & -1 \end{array} \right] \begin{matrix} x_1 \\ x_2 \end{matrix}$$

Testing solution:

$$x_1 \begin{bmatrix} 1 \\ 4 \end{bmatrix} + x_2 \begin{bmatrix} 3 \\ -6 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix} - \begin{bmatrix} 3 \\ -6 \end{bmatrix} = \begin{bmatrix} -2 \\ 10 \end{bmatrix}$$

This shows that $\vec{b}$ is in the column space of A .



$\vec{v}_1$ and $\vec{v}_2$ span $\mathbb{R}^2$, so does $\vec{b}$.

Theorem: A system of linear equations $A\vec{x} = \vec{b}$ is consistent (= solvable), if and only if, $\vec{b}$ is in the column space of A .

*The column space of a Matrix $A = A_{m \times n}$ is a subspace of $\mathbb{R}^m$ and a basis for this subspace consists of columns in $A_{m \times n}$ corresponding to leading-1 columns in the reduced (Gauss) matrix.

3800 Task with basis for a column space, Video

Theorem

* A system of linear equations $A\vec{x} = \vec{b}$ is consistent if, and only if $\vec{b}$ is in the column space of A .

* The column space of a matrix $A = A_{m \times n}$ is a subspace of $\mathbb{R}^m$, and a basis for this subspace consists of columns in $A_{m \times n}$ corresponding to the leading-1-columns in the reduced (Gauss) matrix

Example

Which of the columns in B give a basis for $\text{Col } B$?

$$B = \begin{bmatrix} 1 & 4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Find a basis for $\text{Col } A$ where

$$A = \begin{bmatrix} 1 & 4 & 0 & 2 & -1 \\ 3 & 12 & 1 & 5 & 5 \\ 2 & 8 & 1 & 3 & 2 \\ 5 & 20 & 2 & 8 & 8 \end{bmatrix}$$

$$\begin{array}{l} R_2 \leftarrow R_2 - 3R_1 \\ R_3 \leftarrow R_3 - 2R_1 \\ R_4 \leftarrow R_4 - 5R_1 \end{array} \begin{bmatrix} 1 & 4 & 0 & 2 & -1 \\ 0 & 0 & 1 & -1 & 8 \\ 0 & 0 & 1 & -1 & 4 \\ 0 & 0 & 2 & -2 & 13 \end{bmatrix}$$

$$R_3 \leftarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & 4 & 0 & 2 & -1 \\ 0 & 0 & 1 & -1 & 8 \\ 0 & 0 & 0 & 0 & -4 \\ 0 & 0 & 2 & -2 & 13 \end{bmatrix}$$

$$R_4 \leftarrow R_4 - 2R_2$$

$$\begin{bmatrix} 1 & 4 & 0 & 2 & -1 \\ 0 & 0 & 1 & -1 & 8 \\ 0 & 0 & 0 & 0 & -4 \\ 0 & 0 & 0 & 0 & -3 \end{bmatrix}$$

$$\begin{array}{l} R_2 \leftarrow R_2 + 2R_3 \\ R_3 \leftarrow R_3 \cdot \frac{1}{4} \\ R_4 \leftarrow R_4 + 3R_3 \end{array} \begin{bmatrix} 1 & 4 & 0 & 2 & -1 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

3800 Video Proof of nullspace being a subspace

$$A_{m \times n} = \begin{bmatrix} \vec{a}_{11} & \dots & \vec{a}_{1n} \\ \vdots & & \vdots \\ \vec{a}_{m1} & \dots & \vec{a}_{mn} \end{bmatrix} \begin{matrix} \vec{r}_1 \\ \vdots \\ \vec{r}_m \end{matrix}$$

$\vec{v}_1 \qquad \qquad \vec{v}_n$

Column space of $A_{m \times n}$, $\text{Col } A_{m \times n}$, is the set of all linear combinations of columns in $A_{m \times n}$

$$\text{Col } A_{m \times n} = \text{Span}\{\vec{v}_1, \dots, \vec{v}_n\}$$

Requirements for null space vectors:

- I/ $\vec{0} \in H$
- II/ $\vec{u}, \vec{v} \in H \Rightarrow \vec{u} + \vec{v} \in H$
- III/ $c \in \mathbb{R}, \vec{u} \in H \Rightarrow c\vec{u} \in H$

The null space of a matrix $A_{m \times n}$ is the set $\text{Nul } A_{m \times n}$ of all solutions of the homogeneous equation $A\vec{x} = \vec{0}$.

For a matrix $A_{m \times n}$, $\vec{x} \in \mathbb{R}^n$

Theorem 12

The null space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^n$. Equivalently, the set of all solutions of a system $A\vec{x} = \vec{0}$ of m homogeneous linear equations in n unknowns is a subspace of $\mathbb{R}^n$.

Proof

- I/ $\vec{0} \in H$ because $A\vec{0} = \vec{0}$
- II/ Let $A\vec{u} = \vec{0}$, $A\vec{v} = \vec{0}$, then $A(\vec{u} + \vec{v}) = A\vec{u} + A\vec{v} = \vec{0} + \vec{0} = \vec{0}$
- III/ Let $A\vec{u} = \vec{0}$, $c \in \mathbb{R}$, then $c(A\vec{u}) = c\vec{0} = \vec{0}$.

qed
quod
erat
demonstrandum

Central idea: $\text{Nul}(A) = \{ \vec{x} : A\vec{x} = \vec{0} \}$

The null space of A is the set of all input vectors that the matrix A sends to the zero vector.

The null space of a matrix A is the set $\text{Nul } A$ of all solutions of the homogeneous equation $A\vec{x} = \vec{0}$

$$A = \begin{bmatrix} 3 & 1 & 1 & 6 \\ 1 & -2 & 5 & -5 \\ 4 & 1 & 2 & 7 \end{bmatrix} \text{ Row reduction gives } \begin{matrix} x_1 & x_2 & x_3 & x_4 \\ \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$A\vec{x} = \vec{0} \Rightarrow \begin{bmatrix} 3 & 1 & 1 & 6 \\ 1 & -2 & 5 & -5 \\ 4 & 1 & 2 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ solves the base for null space of } A.$$

From row reduction we have: x_3 and x_4 are free variables. $x_3 = s$, $x_4 = t$.

$$x_1 = -s - t$$

$$x_2 = 2s - 3t$$

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -s-t \\ 2s-3t \\ s \\ t \end{bmatrix} \Rightarrow s \begin{bmatrix} -1 \\ 2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ -3 \\ 0 \\ 1 \end{bmatrix}$$

The set $S = \text{Span}\{\vec{u}, \vec{v}\}$ forms a basis for $\text{Nul } A$.

Let's test it:

$$\left. \begin{aligned} & -1 \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix} + 0 \begin{bmatrix} 6 \\ -5 \\ 7 \end{bmatrix} - 1 \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix} - 3 \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} + 0 \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix} + 1 \begin{bmatrix} 6 \\ -5 \\ 7 \end{bmatrix} \\ & = \begin{bmatrix} -3 \\ -1 \\ -4 \end{bmatrix} + \begin{bmatrix} 2 \\ -4 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix} + \begin{bmatrix} -3 \\ -1 \\ -4 \end{bmatrix} + \begin{bmatrix} -3 \\ 6 \\ -3 \end{bmatrix} + \begin{bmatrix} 6 \\ -5 \\ 7 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \end{aligned} \right\} s, t \in \mathbb{R}$$

Rank = 2 (Rank(A) = 2)
nullity = 2 (Nullity(A) = 2)
Rank = pivot columns
nullity = free variables

Video 3800 Example both Null and Column Space

$$A = \begin{bmatrix} 3 & 1 & 1 & 6 \\ 1 & -2 & 5 & -5 \\ 4 & 1 & 2 & 7 \end{bmatrix}$$

3×4

a) Find a basis for the null space of the system

$$A\vec{x} = \vec{0}$$

b) Find a basis for the column space of A.

Solution:

$$a) A\vec{x} = \vec{0} \Rightarrow \begin{bmatrix} 3 & 1 & 1 & 6 \\ 1 & -2 & 5 & -5 \\ 4 & 1 & 2 & 7 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \vec{0}$$

$A \quad \vec{x} \quad \vec{0}$

$$\begin{bmatrix} 3 & 1 & 1 & 6 \\ 1 & -2 & 5 & -5 \\ 4 & 1 & 2 & 7 \end{bmatrix}$$

$$R_3 \leftarrow R_3 - R_1$$

$$\begin{bmatrix} 3 & 1 & 1 & 6 \\ 1 & -2 & 5 & -5 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

$$R_2 \leftarrow R_2 - R_3$$

$$\begin{bmatrix} 3 & 1 & 1 & 6 \\ 0 & -2 & 4 & -6 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

$$R_2 \leftarrow R_2 \cdot (-1/2)$$

$$\begin{bmatrix} 3 & 1 & 1 & 6 \\ 0 & 1 & -2 & 3 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

$$R_1 \leftarrow R_1 - 3R_3$$

$$\begin{bmatrix} 0 & 1 & -2 & 3 \\ 0 & 1 & -2 & 3 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

$$R_1 \leftarrow R_3$$

$$R_3 \leftarrow R_3 - R_2$$

leading 1's

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

free variables

a) $A\vec{x} = \vec{0}$ solves the base for null space of A.

$$\begin{bmatrix} x_1 & x_2 & x_3 & x_4 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & -2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_3 = s, x_4 = t$$

$$x_1 = -s - t$$

$$x_2 = 2s - 3t$$

b)

$$A \sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -s-t \\ 2s-3t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} -1 \\ 2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ -3 \\ 0 \\ 1 \end{bmatrix}$$

$S = \text{span}\{\vec{u}, \vec{v}\}$
Basis for Null A